WORKED SOLUTION

STEP II 1993, Question 10

Pure • Quadratic Forms and Conics

STEP II 1993, Question 10

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/93-s2-q10

The key idea

A quadratic equation in x and y is messy for two reasons: the xy cross term and the linear terms. The matrix $\mathbf{P}$ deals with the first, because $\mathbf{PAP}$ is diagonal. The shift $\mathbf{a}$ deals with the second: we choose it to move the origin to the centre of the conic, so that every term linear in $\mathbf{x}_1$ cancels. What's left is a pure "sum of squares = constant".

Solution

Diagonalising

First,

$$\mathbf{AP} = \begin{pmatrix} -1 & 8 \\ 8 & 11 \end{pmatrix}\begin{pmatrix} 1 & 2 \\ 2 & -1 \end{pmatrix} = \begin{pmatrix} 15 & -10 \\ 30 & 5 \end{pmatrix},$$

so

$$\mathbf{PAP} = \begin{pmatrix} 1 & 2 \\ 2 & -1 \end{pmatrix}\begin{pmatrix} 15 & -10 \\ 30 & 5 \end{pmatrix} = \begin{pmatrix} 75 & 0 \\ 0 & -25 \end{pmatrix},$$

which is diagonal, as required. Note also that $\mathbf{P}$ is symmetric, so $\mathbf{P}^{\mathrm{T}} = \mathbf{P}$. We'll use this in a moment.

Substituting

Put $\mathbf{x} = \mathbf{P}\mathbf{x}_1 + \mathbf{a}$. Then $\mathbf{x}^{\mathrm{T}} = \mathbf{x}_1^{\mathrm{T}}\mathbf{P} + \mathbf{a}^{\mathrm{T}}$, and

$$\begin{aligned}\mathbf{x}^{\mathrm{T}}\mathbf{A}\mathbf{x} &= \mathbf{x}_1^{\mathrm{T}}\mathbf{PAP}\mathbf{x}_1 + \mathbf{x}_1^{\mathrm{T}}\mathbf{PA}\mathbf{a} + \mathbf{a}^{\mathrm{T}}\mathbf{AP}\mathbf{x}_1 + \mathbf{a}^{\mathrm{T}}\mathbf{A}\mathbf{a} \\ &= \mathbf{x}_1^{\mathrm{T}}\mathbf{PAP}\mathbf{x}_1 + 2\mathbf{a}^{\mathrm{T}}\mathbf{AP}\mathbf{x}_1 + \mathbf{a}^{\mathrm{T}}\mathbf{A}\mathbf{a}.\end{aligned}$$

The two middle terms are equal because each is a 1×1 matrix, so equals its own transpose, and $(\mathbf{x}_1^{\mathrm{T}}\mathbf{PA}\mathbf{a})^{\mathrm{T}} = \mathbf{a}^{\mathrm{T}}\mathbf{A}^{\mathrm{T}}\mathbf{P}^{\mathrm{T}}\mathbf{x}_1 = \mathbf{a}^{\mathrm{T}}\mathbf{AP}\mathbf{x}_1$, using that $\mathbf{A}$ and $\mathbf{P}$ are symmetric. Also $\mathbf{b}^{\mathrm{T}}\mathbf{x} = \mathbf{b}^{\mathrm{T}}\mathbf{P}\mathbf{x}_1 + \mathbf{b}^{\mathrm{T}}\mathbf{a}$. So the equation becomes

$$\mathbf{x}_1^{\mathrm{T}}\mathbf{PAP}\mathbf{x}_1 + (2\mathbf{a}^{\mathrm{T}}\mathbf{A} + \mathbf{b}^{\mathrm{T}})\mathbf{P}\mathbf{x}_1 + (\mathbf{a}^{\mathrm{T}}\mathbf{A}\mathbf{a} + \mathbf{b}^{\mathrm{T}}\mathbf{a} - 11) = 0.$$

Ask yourself What does "a suitable vector $\mathbf{a}$" have to do?

The target $3x_1^2 - y_1^2 = c$ has no terms in x_1 or y_1 alone, so the middle bracket must vanish. Since $\mathbf{A}$ is symmetric, $2\mathbf{a}^{\mathrm{T}}\mathbf{A} + \mathbf{b}^{\mathrm{T}} = (2\mathbf{A}\mathbf{a} + \mathbf{b})^{\mathrm{T}}$, so we need

$$\mathbf{A}\mathbf{a} = -\tfrac{1}{2}\mathbf{b} = \begin{pmatrix} -9 \\ -3 \end{pmatrix}.$$

Here $\det \mathbf{A} = -11 - 64 = -75 \neq 0$, so there is exactly one solution:

$$\mathbf{a} = -\frac{1}{75}\begin{pmatrix} 11 & -8 \\ -8 & -1 \end{pmatrix}\begin{pmatrix} -9 \\ -3 \end{pmatrix} = -\frac{1}{75}\begin{pmatrix} -75 \\ 75 \end{pmatrix} = \begin{pmatrix} 1 \\ -1 \end{pmatrix}.$$

(Check: $\mathbf{A}\mathbf{a} = \begin{pmatrix} -1-8 \\ 8-11 \end{pmatrix} = \begin{pmatrix} -9 \\ -3 \end{pmatrix}$.)

The constant

With this $\mathbf{a}$, we have $\mathbf{a}^{\mathrm{T}}\mathbf{A}\mathbf{a} = \mathbf{a}^{\mathrm{T}}(-\tfrac{1}{2}\mathbf{b}) = -\tfrac{1}{2}\mathbf{b}^{\mathrm{T}}\mathbf{a}$, so the constant term is

$$\tfrac{1}{2}\mathbf{b}^{\mathrm{T}}\mathbf{a} - 11 = \tfrac{1}{2}(18 - 6) - 11 = -5.$$

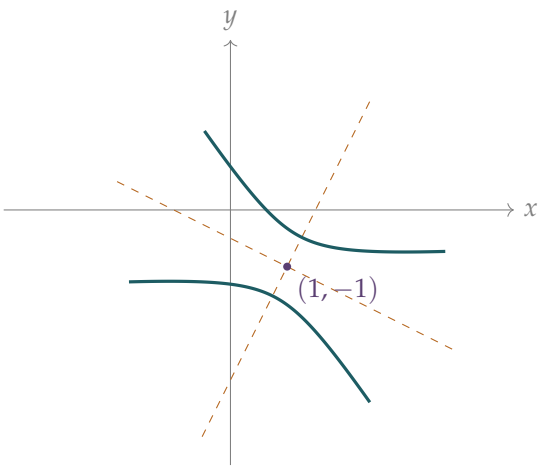
The quadratic term is

$$\mathbf{x}_1^{\mathrm{T}}\begin{pmatrix} 75 & 0 \\ 0 & -25 \end{pmatrix}\mathbf{x}_1 = 75x_1^2 - 25y_1^2.$$

So the equation becomes $75x_1^2 - 25y_1^2 - 5 = 0$. Dividing by 25,

$$3x_1^2 - y_1^2 = \tfrac{1}{5}, \qquad \text{so } c = \tfrac{1}{5}.$$

Written out, the substitution is $x = x_1 + 2y_1 + 1$, $y = 2x_1 - y_1 - 1$. The original equation is $-x^2 + 16xy + 11y^2 + 18x + 6y - 11 = 0$, so the curve is a hyperbola with centre $(1, -1)$. The columns of $\mathbf{P}$, $(1, 2)$ and $(2, -1)$, are perpendicular, so they give its axes of symmetry.



The dashed lines are the axes of symmetry, in the directions $(1,2)$ and $(2,-1)$.

Ask yourself Why isn't $\mathbf{P}$ a rotation?

Its columns are perpendicular, but each has length $\sqrt{5}$, so $\mathbf{P}^2 = 5\mathbf{I}$ rather than $\mathbf{I}$. That's why the diagonal entries of $\mathbf{PAP}$ are 75 and -25: they are 5 times the eigenvalues 15 and -5 of $\mathbf{A}$. The scale factor doesn't matter here, because it just multiplies the quadratic part by 5.

Checks

- The trace and determinant of $\mathbf{A}$ are 10 and -75, so its eigenvalues are 15 and -5. These match $75/5$ and $-25/5$.
- Take $x_1 = y_1 = 0$. Then $(x,y) = (1,-1)$ and the original left-hand side is $-1-16+11+18-6-11 = -5$, which agrees with $75x_1^2 - 25y_1^2 - 5$.
- Take $x_1 = 1$, $y_1 = 1$. Then $(x,y) = (4,0)$, and $-16+72-11 = 45 = 75-25-5$.

We also checked the identity LHS $= 25\left(3x_1^2 - y_1^2 - \frac{1}{5}\right)$ exactly at 1000 random rational points.

Common slips

- Forgetting that the cross terms $\mathbf{x}_1^{\mathrm{T}}\mathbf{PAa}$ and $\mathbf{a}^{\mathrm{T}}\mathbf{APx}_1$ are equal, which loses the factor 2 and gives $\mathbf{Aa} = -\mathbf{b}$.
- Stopping at $75x_1^2 - 25y_1^2 = 5$ and giving $c = 5$, without dividing through to match the required form.

Beyond STEP

This is the principal axes theorem in miniature. Every real symmetric matrix can be diagonalised by an orthogonal matrix (here $\mathbf{P}/\sqrt{5}$). So every conic, and every quadric surface in three dimensions, can be put into standard form by a rotation (or reflection) and a translation. The same idea classifies stationary points of functions of several variables through the Hessian matrix. Strang's *Introduction to Linear Algebra* has a good chapter on symmetric matrices and positive definiteness.

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